

Praneeth Tadepalli

tadep013@umn.edu | (763) 321-7907 | linkedin.com/in/PraneethTadepalli

Education

University of Minnesota Twin Cities, MN Sep 2025 – May 2027

MS in Electrical and Computer Engineering | GPA: 3.778/4.0

- **Coursework:** Analog IC Design(Weak Inversion, gm/Id methodology), VLSI Design I(Combinational and sequential Circuit Design, FinFETs, Delay Optimization), VLSI Design Lab(RTL-to-GDSII flow)

IITDM Kancheepuram, Chennai, India Dec 2021 – May 2025

B.Tech in Electronics and Communication Engineering | CGPA: 8.98/10.0

- **Coursework:** Analog Circuits, Control Systems, Signals and Systems, Communication Systems

Technical Skills

Programming: MATLAB, C, Python, Embedded C, TCL

HDL & Modeling: Verilog, Verilog A, System Verilog

EDA Tools: LTSpice, Hspice, Cadence Virtuoso, Spectre, Calibre, NI LabView, NI Multisim, Synopsys DV, ICC2, Prime Time

Experience

Mobile & Cloud Computing Laboratory - University of Hyderabad, India May 2024 - Sep 2024

Research Intern

- Designed a solar-powered charging circuit for *Raspberry Pi 4* using a solar panel, battery pack, and charge controller with voltage regulation
- Integrated safety components including capacitor, polyfuse, and switch for reliable and protected power delivery

Projects

Low-Power Wide Swing Cascode OTA with Switched-Capacitor Biquad Filter for Sep 2025 - Dec 2025

ECG Signal Amplification | Cadence Virtuoso, Spectre

- Designed a Differential Wide Swing Cascode OTA using *CMOS 50 nm* technology, focusing on low power and compact area with a gain of *51.1 dB* and Unity Gain Frequency of *128.8 MHz*
- Validated the performance across *PVT* variations with an FOM ranging from *1175 to 1380*
- Developed a high Q switched-capacitor biquad filter for low-frequency ECG signal filtering and amplification with a *Q factor of 1.037*

Markov Chain Monte Carlo (MCMC) Processor with Knuth–Yao Discrete Sampler Sep 2025 - Dec 2025

Core | Synopsys DV, Prime Time, Synopsys ICC2

- Designed a Verilog-based Knuth–Yao discrete sampler core utilizing an LFSR-based random number generator (RNG) for probabilistic sampling in MCMC processors
- Implemented and verified the RTL architecture using Synopsys Design Compiler and VCS, with focus on timing, area, and power optimization
- Completed RTL-to-GDSII flow including synthesis, floor-planning, placement & routing, and sign-off analysis

8-bit Threshold Leaky ReLU (TLReLU) | Cadence Virtuoso, Hspice Oct 2025 - Dec 2025

- Designed an 8-bit TLReLU circuit in *TSMC 16 nm* CMOS technology using a carry-select adder architecture to minimize the critical path delay
- The optimized design results in a total transistor count of *1098* and a propagation delay of *120 ps*

Inverter Layout Design using | Cadence Virtuoso, Calibre Sep 2025 - Oct 2025

- Designed and verified a 4-fin CMOS inverter layout in *TSMC 16 nm* technology, achieving *DRC and LVS* clean results
- Performed advanced verification by creating and validating a *3×3* grid inverter layout achieving *DRC* clean results

Analog Spiking Neural Network | Cadence Virtuoso, Spectre Jun 2024 - May 2025

- Simulated neural circuits including *Axon Hillock*, *Voltage Amplifier I&F*, *Low-Power I&F*, and *AdEx* models in Cadence Virtuoso using *gpdk 90 nm* technology
- Implemented analog synaptic plasticity learning rules using *Pair-based* and *Triplet-based STDP* circuits with different memristor models
- Developed a *BCM-based ECG signal classifier* using Triplet STDP with a memristor circuit, achieving *98%* accuracy